

Total No. of Questions : 4]

SEAT No. :

PB260

[Total No. of Pages : 2

[6270]-48

B.E. (Civil Engineering) (Insem)

GREEN STRUCTURES AND SMART CITIES

(2019 Pattern) (Semester - VIII) (Elective - VI) (401014E)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.1 or Q.2 or Q.3 or Q.4.
- 2) Figures to the right indicate full marks.
- 3) Your answers will be valued as a whole.
- 4) Assume suitable data, if necessary.
- 5) Use of electronic pocket calculator and steam table is allowed.

Q1) Answer the following-

- a) Give note on [5]
 - i) Life cycle energy analysis
 - ii) Life cycle assessment (LCA)
- b) How the use of local materials contribute to minimize the embodied energy? Give suitable example. [5]
- c) How the bio-capacity can be increase? Discuss some strategies involved in it. [5]

OR

Q2) Answer the following-

- a) How can building design and construction practices be changed to minimize embodied energy in future construction projects? [5]
- b) Discuss the relationship between economic development and ecological footprint. [5]
- c) How might climate change impact bio-capacity? Discuss with suitable example. [5]

P.T.O.

Q3) Answer the following-

- a) Enlist and discuss certification programs for green building. What is the importance of these certification in green building? [5]
- b) What are the different strategies for operational energy reduction in buildings? Give suitable example. [5]
- c) How the renewable energy sources that can be integrated into buildings to achieve net zero? Discuss. [5]

OR

Q4) Answer the following-

- a) What are the different options to be considered during building design phase to make building energy-efficient? [5]
- b) Discuss the radiation budget. What are its components? [5]
- c) How green spaces mitigate the air pollution? Enlist the factors be considered while selecting tree species for green spaces. [5]